Data Communications

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Data Communications (A18121)

Short Title:	Data Communications
Faculty	Science and Computing
Department	Engineering Technology
Cluster	Data Communications
Author	POLEARY
Credits	5
Level:	Advanced

Description of Module / Aims

This module complements the Telecommunications taught in the course by introducing the structures that lie behind the transmission and reception of a communications signal. The course examines how, after the recovery of the transmitted bit/symbol, how the original transmitted information may be interpreted in a structured system in order to effectively communicate, how to manage errors and also how to secure the communication.

Programmes

`COME-0005`	BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	4 / 7 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	4 / 7 / M
`COME-0005`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	4 / 7 / E
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	4 / 7 / M
`COME-0005`	BEng (Hons) (`WD_ETRIC_B`)	4 / 7 / M
`DCOM-0001`	BSc (Hons) in Applied Electronics (`WD_EELEC_B`)	2 / 3 / M

Indicative Content

- 1 OSI and Internet Models for communication
- 2 Physical Layer (Modulation, clock recovery, framing, sunchronisation and flags)
- 3 Data Link Layer (Including flow and error control)
- 4 Network Layer (network algorithms)
- 5 Internet Protocol (datagram structures, constructng a header field for IPv4 and IPv6)
- 6 Transport Layer (role, implementations in TCP and UDP)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Critically analyse (de)modulation, clock and data recovery block diagrams.
2. Describe the implementation of flow and error control in general and develop & test a Cyclic Redundancy Check error detection system.
3. Examine a data frame or packet and prepare evidence-based conclusions to evaluate the nature of the underlying datagram or switched communication protocol..
4. Assess the requirements of a Transport Layer protocol, when the underlying network protocol is unpredicable in terms of losses and delays.
5. Demonstrate a practical ability to develop, troubleshoot, test, analyse and report on the Data Communications topics covered here.
6. Select a means of securing communications and apply advanced skills to develop a bespoke system.

Learning and Teaching Methods

- In-class: Lectures, Group Discussion, Practical exercises
- Online: Lecture notes, Laboratory descriptions, Assignments; Discussion forum driven by lecturer
- Laboratory exercises

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3, 4, 6
Continuous Assessment	50%	1, 2, 3, 4, 5, 6
Practical	30%	2, 5, 6
Assignment	15%	5
Participation	5%	6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Lab	12	
Independent Learning	87	

Supplementary Material

- Stallings, W. _Data and computer communications_. New Jersey: Prentice Hall, 2014.